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Why even have an SDK?

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whoami - Samuel Colvin

- Creator of **Pydantic**
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The Pydantic Stack

- **Pydantic Validation**: contracts for data
- **Pydantic AI**: production grade agent framework
- **Pydantic Logfire**: observability, evals, prompt management, LLMOps
- **Pydantic AI Gateway**: one endpoint, every model
- **Pydantic Monty**: sandboxed Python interpreter

**OpenTelemetry already exists.
Coding agents now exist.**

So why did we write three SDKs and a CLI?

What OpenTelemetry gives you

- **A wire format** - OTLP, plus semantic conventions for what to call things
- **Instrumentation for everything** - `opentelemetry-instrumentation-fastapi`, `-httpx`, `-asyncpg`, ~100 more
- **Any backend** - Jaeger, Tempo, Datadog, Logfire, a file

What it doesn't give you

- **An opinion.** A tracer, a span and an attribute bag. The rest is up to you.
- Setting it up by hand is ~40 lines of provider, processor, exporter, resource
- And nothing in the spec says "show me this trace *while it runs*"

The SDK is thin. On purpose.

```
# logfire-sdk/pyproject.toml
dependencies = [
    "opentelemetry-sdk",
    "opentelemetry-exporter-otlp-proto-http",
    "opentelemetry-instrumentation",
    "rich",
    "requests",
    "protobuf",
    "typing-extensions",
    "executing",
]
```

- Three OTEL packages, a console renderer, and `executing` (more on that later)
- **No pydantic.** The Pydantic observability SDK does not depend on Pydantic
- 31 extras, each one an `opentelemetry-instrumentation-*` package
- `logfire-api`: a zero-dependency no-op shim with the same API, for libraries
- `send_to_logfire=False` + `additional_span_processors=[...]` sends to any OTLP backend, using plain OTEL types

Two lines

```
import logfire
from fastapi import FastAPI

app = FastAPI()

logfire.configure()
logfire.instrument_fastapi(app)
```

- `instrument_fastapi` is `opentelemetry-instrumentation-fastapi` with our defaults, our message formatting, and our span tagging on top
- 34 `instrument_*` one-liners, all built the same way: Pydantic AI, OpenAI, Anthropic, httpx, asynccpg, Django, Celery, Redis, MCP, Monty, ...
- `logfire.configure()` reads `.logfire/`, env vars, or the token you pass - and works with no arguments

**The SDK carries opinions
the backend depends on.**

templates · schemas · pending spans

Opinion 1: the template is the name

```
logfire.info('hello {name}', name=user)

# span name           'hello {name}'
# logfire.msg_template 'hello {name}'
# logfire.msg         "hello name=User(id=1, email='x@y.z', signed_up=datetime(...))"
# logfire.json_schema  {"properties": {"name": {"x-python-datatype": "PydanticModel", ...}}}
```

- The **template** is low cardinality: group, count, alert on it
- The **message** is rendered for humans, values included
- f-strings work too: `logfire.info(f'hello {user.email}')` - we recover the template from the AST with `executing`
- Attributes ride along as ordinary OTel attributes; the schema only carries what a JSON blob would lose

Opinion 2: values look like Python

What the wire carries

```
{"id": 1, "email": "x@y.z",  
  "signed_up": "2026-01-01T09:30:00Z"}
```

plus, once per attribute, in `logfire.json_schema` :

```
{"title": "User",  
  "x-python-datatype": "PydanticModel",  
  "properties": {"signed_up": {  
    "type": "string", "format": "date-time",  
    "x-python-datatype": "datetime"}}}
```

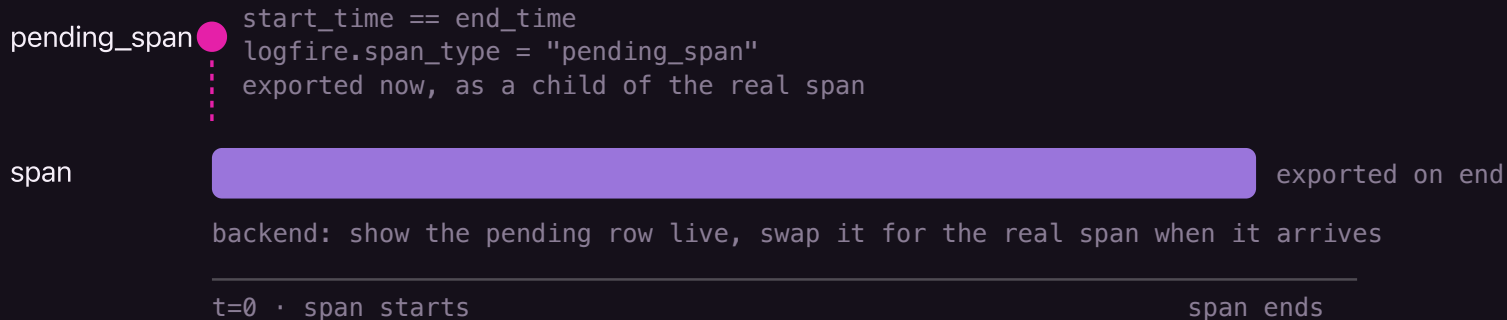
What you see

```
User(  
    id=1,  
    email='x@y.z',  
    signed_up=datetime(2026, 1, 1, 9, 30),  
)
```

- `x-python-datatype` for models, dataclasses, `datetime`, `Decimal`, `UUID`, `Enum`, `Path`, exceptions
- DataFrames carry `x-columns`, `x-row-count`, `x-shape`
- Plain `str`, `int`, `float`, `bool` need nothing: `{}`

Opinion 3: pending spans

- OTLP has no "span started" message. Exporters run on `on_end`, so a slow agent run is invisible until it finishes
- We emit a **zero-duration twin** when a span starts, tagged so the backend can reconcile it later



- `RemovePendingSpansExporter` drops the twin when the real span lands in the same batch, so fast spans cost nothing
- Only the Logfire exporter sees them: your `additional_span_processors` never do

Same opinions, three languages

	PYTHON LOGFIRE-SDK 6	JS / TS LOGFIRE 0.9	RUST LOGFIRE 0.12
Built on	opentelemetry-sdk	@opentelemetry/sdk-node	tracing + opentelemetry
Auto-instrumentation	34 instrument_* (OTel contrib)	node auto-instrumentations	anything using tracing
logfire.msg + msg_template	✓	✓	msg; template = span name
logfire.json_schema	✓	✓	✓
Pending spans	✓	✗	✓
Prompts / variables	✓	OpenFeature / OFREP	OpenFeature / OFREP

Hello, TypeScript

```
import * as logfire from 'logfire'

logfire.configure({ serviceName: 'weather-app' })

logfire.span('forecast for {city}', { city: 'Paris', units: 'C' }, {}, async () => {
  const forecast = await fetchForecast('Paris')
  logfire.info('got {days} days', { days: forecast.length, forecast })
})
```

- `npm i logfire` - wraps `@opentelemetry/sdk-node` and Node auto-instrumentations
- `@pydantic/logfire-api` (zero deps, for libraries), `-browser`, `-cf-workers`
- Same `logfire.msg_template` / `logfire.msg` / `logfire.json_schema` attributes

Hello, Rust

```
fn main() -> Result<(), Box<dyn std::error::Error>> {  
    let logfire = logfire::configure().finish()?;  
    let _guard = logfire.shutdown_guard();  
  
    logfire::span!("forecast for {city}", city = "Paris").in_scope(|| {  
        logfire::info!("got {days} days", days = 7u64);  
    });  
    Ok(())  
}
```

- `cargo add logfire` - built on `tracing` and `opentelemetry`, so anything already instrumented with `tracing` just works
- Pending spans included; `logfire.msg` and `logfire.json_schema` are set by the macros
- It's the same crate the CLI is built on

**Agents don't make the SDK redundant.
They make it the interface.**

**A small, stable, discoverable surface is exactly what an agent
can learn.**

Demo: one trace, from agent to app

```
logfire.configure(service_name='weather-app')
logfire.instrument_pydantic_ai()
logfire.instrument_monty()

app = FastAPI()
logfire.instrument_fastapi(app)

@app.get('/weather')
async def weather(q: str) -> dict[str, str]:
    result = await weather_agent.run(f"What's the weather in {q}, in Celsius?")
    return {'answer': result.output}
```

- HTTP request → Pydantic AI run → LLM call → Monty codemode execution → tool calls, one trace
- Pending spans: the trace is on screen before the model has answered

Prompts are variables

```
prompt = logfire.template_var(  
    name='prompt__weather_agent',  
    type=str,  
    default='Combine tool calls into a single code execution call. Answer in {{units}}.',  
    inputs_type=WeatherInputs, # a BaseModel with `units: str`  
)  
  
with prompt.get(WeatherInputs(units='celsius'), label='production') as resolved:  
    result = await weather_agent.run(question, instructions=resolved.value)
```

- A prompt is a managed variable named `prompt__<slug>`: versioned, targeted, rolled out like a feature flag
- Handlebars `{{placeholders}}`, and `@{other_variable}@` composition
- `logfire[variables]` extra in Python. Everyone else: Logfire speaks **OFREP**, so any **OpenFeature** client evaluates the same variables

One binary, every ecosystem

```
➤ uv tool install logfire-cli # or: uvx logfire-cli
➤ npm install --global logfire-cli # or: npx logfire-cli
➤ pip install logfire # logfire 6 = logfire-sdk + logfire-cli
```

- One Rust binary. PyPI wheels for 17 targets, npm for 8. No Rust, no Python needed to run it
- **For humans:** `logfire signup`, `auth`, `init`, `project current`, `token read`
- **For scripts:** `--output json|jsonl`, `--no-input`, stable exit categories
- **For agents:** `logfire wizard` hands your repo to Claude Code or Codex with our setup skill; `logfire mcp ...` is the same hosted MCP server your agent uses

Demo: auth + query

```
logfire auth
logfire mcp projects

logfire mcp query run \
  'SELECT span_name, duration, message FROM records ORDER BY start_timestamp DESC LIMIT 20' \
  --project demo

logfire --output json mcp query run --file slow.sql --project demo
```

- The query goes through the hosted MCP server, so a coding agent takes exactly the same path
- Same binary, same auth, same output contract for people, pipelines and agents

Thank you

Thanks to **Modal** for hosting



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